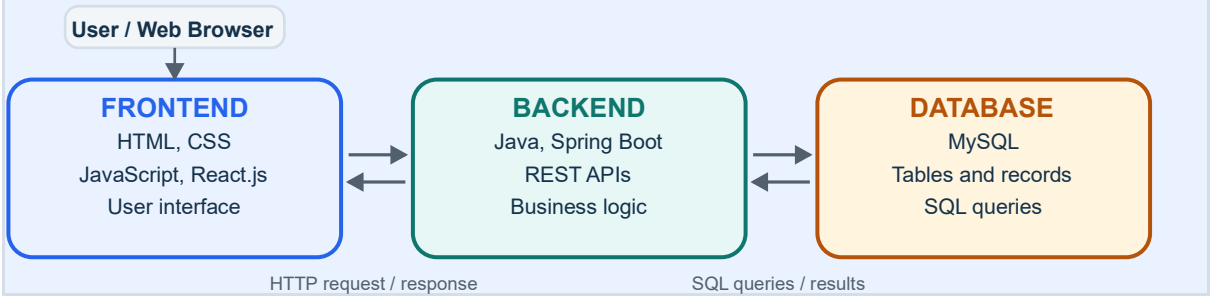
INTRODUCTION TO FULL STACK DEVELOPMENT

TechnoSignia Internship — Day 1

Objective

Understand how the frontend, backend, and database work together to build a web application.



1. What Is Full Stack Development?

Full Stack Development is the development of the different layers of a web application, including the frontend, backend, database, and the communication between them. A Full Stack Developer can work across these layers.

Example: In an online shopping application, the frontend displays products, the backend processes orders and payments, and the database stores product, customer, and order information.

2. Frontend, Backend, and Database

Layer	Responsibility	Technologies / Example
Frontend	The part users see and interact with: pages, forms, buttons, and menus.	HTML, CSS, JavaScript, React.js; a login page.
Backend	Receives requests, validates input, applies business rules, and prepares responses.	Java, Spring Boot, REST APIs; checking login credentials.
Database	Stores, organizes, and retrieves application data.	MySQL and SQL; user accounts and product records.

Easy way to remember: Frontend = what the user sees; Backend = what the application does; Database = what the application stores and retrieves.

3. Client-Server Architecture

A client is a device or application—commonly a web browser—that requests a service. A server receives the request, processes it, and sends a response. In a typical web application, the frontend acts as the client and the backend provides server-side functionality.

Typical request-response flow

1	The user performs an action, such as clicking Login.
2	The frontend sends an HTTP request to a backend endpoint.
3	The backend validates the request and applies business rules.
4	If needed, the backend queries the database and receives results.
5	The backend returns an HTTP response, often containing JSON.
6	The frontend displays the result to the user.

4. Java Full Stack Development

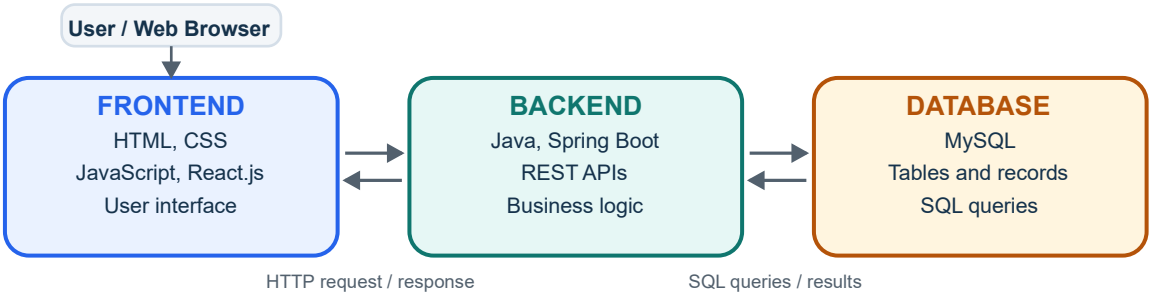
Java Full Stack Development commonly uses Java on the backend and web technologies on the frontend. A typical application may use React.js for the user interface, Spring Boot for backend APIs, and MySQL for persistent data storage.

Technology overview

Technology	Purpose
HTML	Defines the structure and content of web pages.
CSS	Controls visual presentation, layout, and styling.
JavaScript	Adds behavior and interactivity in the browser.
React.js	A JavaScript library for building component-based user interfaces.
Java	A programming language used for backend application logic.
Spring Boot	Simplifies creating Java applications and REST APIs.
SQL	A language for querying and modifying relational database data.
MySQL	A relational database management system.
Git / GitHub	Git tracks code changes; GitHub hosts repositories for collaboration and sharing.
Maven / Postman	Maven manages Java dependencies and builds; Postman helps test APIs.

5. Architecture Diagram

The diagram below summarizes the common communication path. The frontend sends requests to the backend; the backend communicates with the database and returns a response to the frontend.



Typical Java Full Stack flow: the frontend calls the backend; the backend accesses the database.

6. Key Terms

Client: The application or device requesting a service.

Server: A system that receives and processes requests.

API: A defined way for software components to communicate.

REST API: A common approach to designing web APIs using HTTP methods such as GET, POST, PUT, and DELETE.

HTTP: A protocol used for communication between web clients and servers.

JSON: A text format commonly used to exchange structured data between a frontend and backend.

SQL: A language used to query and modify data in relational databases.

7. Summary

A web application commonly separates responsibilities into frontend, backend, and database layers. The frontend provides the user interface, the backend processes requests and applies business rules, and the database stores and retrieves persistent information. In a Java Full Stack application, HTML, CSS, JavaScript, and React.js may be used for the frontend; Java and Spring Boot for the backend; and MySQL with SQL for data storage and queries.